3. Indices and Cube root

(i) $a^m \times a^n = a^{m+n}$

(ii) $a^m \div a^n = a^{m-n}$

(iii) $(a \times b)^m = a^m \times b^m$

(iv) $a^0 = 1$

(v) $a^{-m} = \frac{1}{a^m}$

(vi) $(a^m)^n = a^{mn}$

(vii) $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

(viii) $\left(\frac{a}{b}\right)^{-m} = \left(\frac{b}{a}\right)^m$

(I) Meaning of the numbers when the index is a rational number of the form $\frac{1}{n}$.

---: Practice Set 3.1 :---

1. Express the following numbers in index form.

(1) Fifth root of 13

$\Rightarrow 13^{\frac{1}{5}}$

(2) Sixth root of 9.

$\Rightarrow 9^{\frac{1}{6}}$

(3) Square root of 256

$\Rightarrow 256^{\frac{1}{2}}$

(4) Cube root of 17.

$\Rightarrow 17^{\frac{1}{3}}$

(5) Eighth root of 100

$\Rightarrow 100^{\frac{1}{8}}$

(6) Seventh root of 30.

$\Rightarrow 30^{\frac{1}{7}}$

2. Write in the form 'nth root of a' in each of the following numbers.

(1) $(81)^{\frac{1}{4}}$

$\Rightarrow$ fifth root of 81

(2) $49^{\frac{1}{2}}$

$\Rightarrow$ Square root of 49

(3) $(15)^{\frac{1}{5}}$
$\Rightarrow$ Fifth root of 51.

(4) $(512)^{\frac{1}{9}}$
$\Rightarrow$ Ninth root of 512

(5) $100^{\frac{1}{19}}$
$\Rightarrow$ Nineteenth root of 100

(6) $(6)^{\frac{1}{7}}$
$\Rightarrow$ Seventh root of 6

Practice Set 3.2

1. Complete the following table.

Sr. no	Nos.	Power of the root	Root of the power
(1)	$(225)^{\frac{3}{2}}$	Cube of square root of 225	Square root of cube of 225
(2)	$(45)^{\frac{4}{5}}$	Fourth of fifth root of 45	Fifth root of fourth of 45
(3)	$(81)^{\frac{6}{7}}$	Sixth of seventh root of 81	Seventh root of sixth of 81
(4)	$(100)^{\frac{4}{10}}$	Fourth of tenth root of 100	Tenth root of fourth of 100
(5)	$(21)^{\frac{3}{7}}$	Third of seventh root of 21	Seventh root of third of 21

2. Write the following numbers in the form of rational indices.

(1) Square root of 5^{th} power of 121
$\Rightarrow (121)^{\frac{5}{2}}$

(2) Cube of 4^{th} root of 324
$\Rightarrow (324)^{\frac{3}{4}}$

(3) 5^{th} root of square of 264
$\Rightarrow (264)^{\frac{2}{5}}$

(4) Cube of cube root of 3
$(3)^{\frac{3}{3}}$

Practice Set 3.3

1. Find the cube roots of the following ~~pr~~ numbers.

1) 8000

→ $8000 = 2 \times 4000$

$= 2 \times 2 \times 2000$

$= 2 \times 2 \times 2 \times 1000$

$= 2 \times 2 \times 2 \times 2 \times 500$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 250$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 125$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 25$

$8000 = \underline{2 \times 2 \times 2} \times \underline{2 \times 2 \times 2} \times \underline{5 \times 5 \times 5}$

$= 2 \times 2 \times 5$

$= 20$

$\sqrt[3]{8000} = 20$

2) 729

$\Rightarrow$ $729 = 3 \times 243$

$= 3 \times 3 \times 81$

$= 3 \times 3 \times \cancel{3\times}\ 27$

$= 3 \times 3 \times 3 \times 3 \times 9$

$729 = \underline{3 \times 3 \times 3} \times \underline{3 \times 3 \times 3}$

$729 = 3^3 \times 3^3 = 9$

$\sqrt[3]{729} = 9$

3) 343

$\rightarrow$ $343 = 7 \times 49$

$= 7 \times 7 \times 7$

$343 = 7^3$

$\sqrt[3]{343} = 7$

4) -512

$\rightarrow$ $512 = 2 \times 256$

$= 2 \times 2 \times 128$

$= 2 \times 2 \times 2 \times 64$

$= 2 \times 2 \times 2 \times 2 \times 32$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 16$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 8$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 4$

$= \underline{2 \times 2 \times 2} \times \underline{2 \times 2 \times 2} \times \underline{2 \times 2 \times 2}$

$512 = (2)^3 \times (2)^3 \times (2)^3$

$512 = 8$

$\sqrt[3]{512} = -8$

5) -2744

$\Rightarrow$ $2744 = 2 \times 1372$

$= 2 \times 2 \times 686$

$= 2 \times 2 \times 2 \times 343$

$= 2 \times 2 \times 2 \times 7 \times 49$

$= \underline{2 \times 2 \times 2} \times \underline{7 \times 7 \times 7}$

$-2744 = (2)^3 \times (7)^3$

$-2744 = -14$

$= \sqrt[3]{-2744} = -14$

6) 32768

$\Rightarrow$ $32768 = 2 \times 16384$

$= 2 \times 2 \times 8192$

$= 2 \times 2 \times 2 \times 4096$

$= 2 \times 2 \times 2 \times 2 \times 2048$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 1024$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 512$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 256$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 128$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times \cancel{64}\ 2 \times 64$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 32$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 16$

$= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 8$

$= \underline{2 \times 2 \times 2 \times 2 \times 2 \times 2} \times \cancel{2}\ \underline{2 \times 2 \times 2 \times 2 \times 2 \times 4}$

$\underline{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times \cancel{2}}$

$32768 = (2 \times 2 \times 2 \times 2 \times 2)^3$

$32768 = (32)^3$

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